5 **(a)** State how line spectra, together with the concept of a photon, provide evidence for discrete energy levels in isolated atoms.

.....

.....

.....

..... [2]

(b) An atom has three energy levels as shown in Fig. 5.1.



Fig. 5.1

The atom produces three lines in its emission spectrum but only two lines in its absorption spectrum. The wavelengths of the two absorption lines are 500 nm and 300 nm.

(i) Given that the energy of the ground state E_1 is -7.60×10^{-19} J, determine the energy of the first excited state E_2.

energy = ... J [2]

(ii) Determine the wavelength of the emission line that is present in the emission spectrum but absent from the absorption spectrum.

wavelength = ... m [2]